

The Perceived Fragility of Explanations in Audio Models: Manipulation of Attribution with Unchanged Predictions

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Abstract

Problem: Current explanations of audio deepfakes are fragile and susceptible to manipulation.

Solution: We introduce a novel psychoacoustic framework that optimizes inaudible perturbations to systematically decouple model attributions from classifications.

Key Finding: Adversaries can distort automated explanation heatmaps while preserving both predicted labels and high perceptual audio quality across state-of-the-art architectures.

Introduction

The Stakes: Explainable AI (XAI) is used to build trust in audio deepfake detection by highlighting deceptive acoustic artifacts.

The Security Gap: Current explanation-tampering research relies on vision-centric metrics (L_p norms) that ignore human auditory limits [1].

Core Contributions: First framework adapting explanation attacks specifically to the audio domain. Dynamic psychoacoustic masking that hides adversarial noise under human hearing thresholds. Comprehensive testing on the SONICS deepfake dataset [2].

Methods

1. Target Elements

- **XAI Tools:** Grad-CAM [3] and LRP [4]
- **Models Evaluated:** VGGish (convolutional), AST (Audio Spectrogram Transformer), and SpecTTRa.

2. The Custom Psychoacoustic Attack

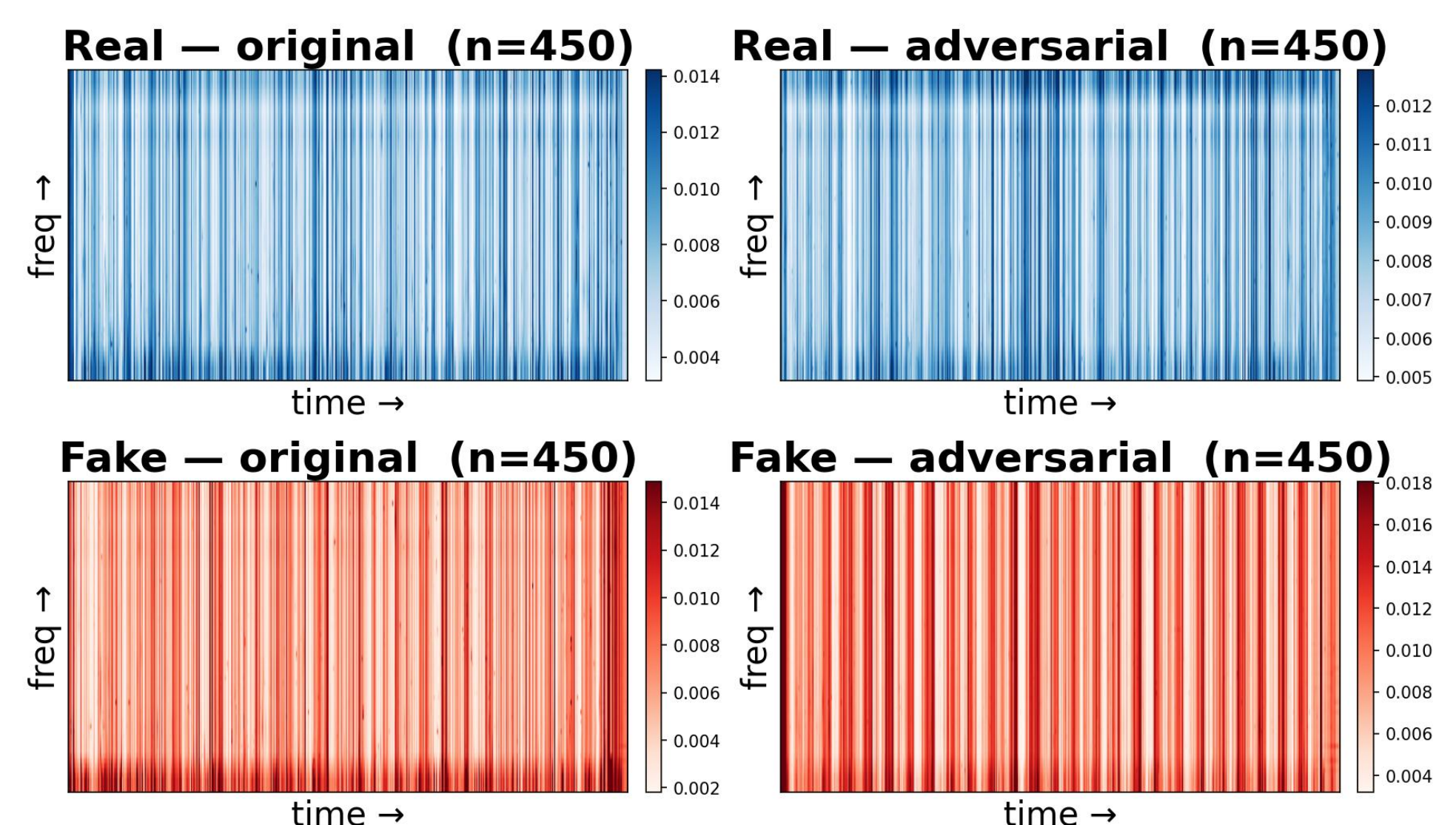
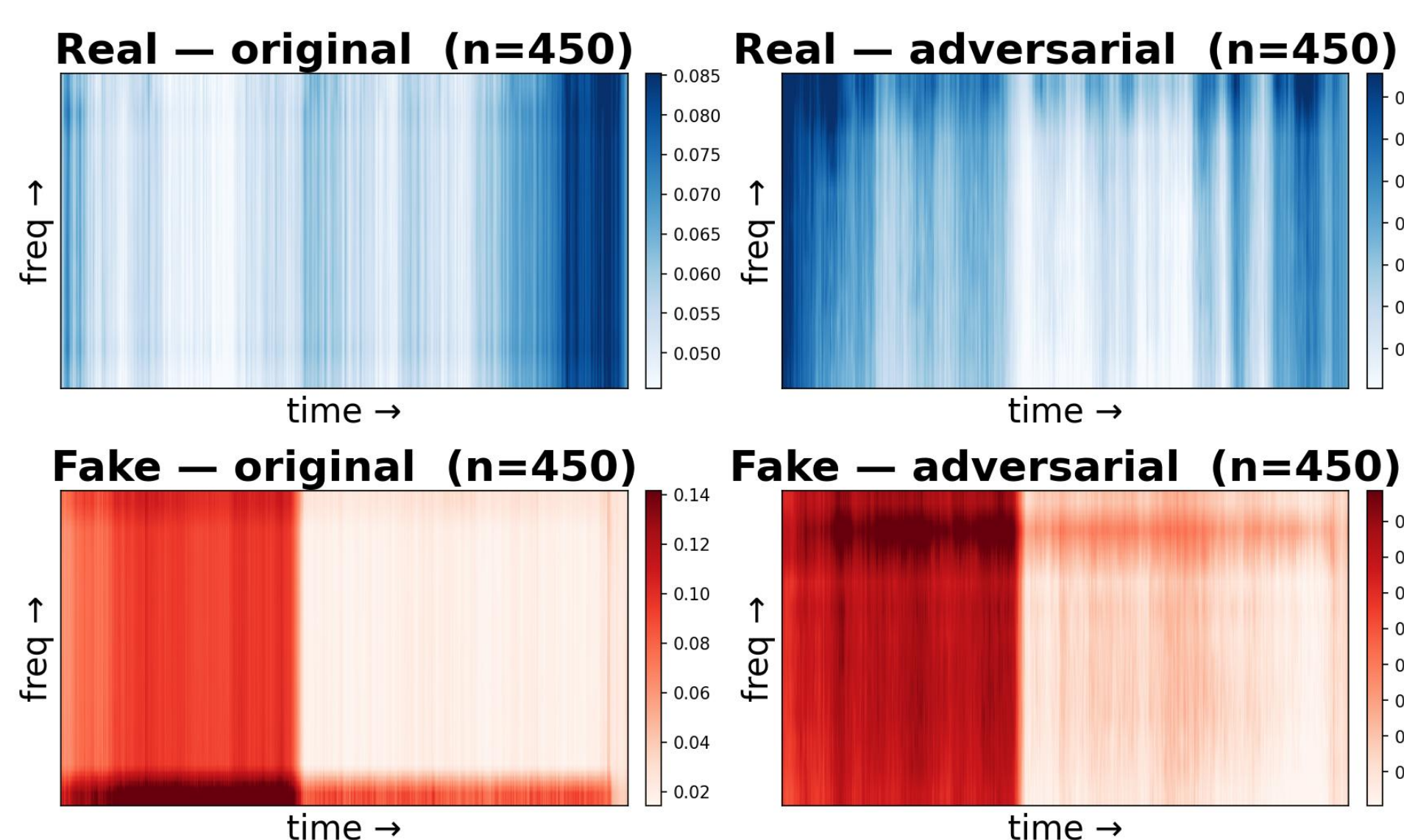
Optimizes a multi-objective loss function to force map displacement while hiding the noise:

$$\mathcal{L}(\delta) = \mathcal{L}_{exp}(\delta) + \lambda_{aud}\mathcal{L}_{aud}(\delta) + \lambda_{pred}\mathcal{L}_{pred}(\delta)$$

$\mathcal{L}_{exp}(\delta)$: Breaks down the cosine similarity of the maps.

$\mathcal{L}_{aud}(\delta)$: Dynamically penalizes noise only if it exceeds the audio's static masking threshold.

$\mathcal{L}_{pred}(\delta)$: Prevents the model's deepfake classification from flipping.



Results

Perceptual Transparency: Standard attacks (PGD) severely degraded audio quality (PESQ $\approx$ 2.8), whereas the **Psychoacoustic Attack** achieved high audio fidelity

(VISQOL $>$ 4.1, CDPAM $\geq$ 0.98).

Vulnerability Ranking: Attention-based models are the most fragile. AST was the easiest to trick (Mean Rank: 3.00 ± 0.58).

SpecTTRa was the most robust (Mean Rank: 7.83 ± 0.48).

Acoustic Blindspots: Dense broadband signals (rock/electronic music) offer larger masking "budgets," making them **easy** to manipulate. Sparse tracks (classical music/silences) drastically restrict the optimizer, making them **hard** to manipulate.

Model	Attack	PESQ $\uparrow$ [-0.5, 4.5]	STOI $\uparrow$ [0, 1]	ViSQOL $\uparrow$ [1, 5]	PEAQ $\uparrow$ [-4, 0]	Zimtohrli $\uparrow$ [0, 5]	CDPAM $\uparrow$ [0, 1]
AST	Psychoacoustic (Ours)	4.06	0.987	4.64	-2.00	4.42	0.989
	PGD	2.77	0.952	3.80	-3.39	3.10	0.858
	X-Shift	3.87	0.990	4.46	-1.80	3.47	0.950
SpecTTRa	Psychoacoustic (Ours)	3.77	0.960	4.15	-2.55	3.79	0.981
	PGD	2.76	0.950	3.79	-3.39	3.14	0.859
	X-Shift	3.74	0.993	4.48	-2.10	3.70	0.925
VGGish	Psychoacoustic (Ours)	4.43	0.997	4.89	-0.41	4.62	0.995
	PGD	2.84	0.953	3.86	-3.37	3.22	0.842
	X-Shift	3.78	0.990	4.31	-2.16	3.56	0.938

Configuration	Median Rank	Mean Rank ($\pm$ SD)
SpecTTRa	8.0	7.83 ± 0.48
VGGish	4.5	4.17 ± 0.95
AST	3.0	3.00 ± 0.58
X-Shift	6.0	5.83 ± 1.28
PGD	5.5	5.00 ± 0.68
Psychoacoustic	3.0	4.17 ± 1.28

Conclusions

The Takeaway: Relying on visual explanation heatmaps as a standalone audit tool for audio AI reliability is dangerous and premature.

Future Work: Interpretability mechanisms must be mathematically bound and locked directly to the classifier's exact decision boundaries, rather than generated post-hoc.

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Reference

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